Satellite Communication

Introduction to Satellite Communication

Introduction to Satellite Communication

What is satellite?

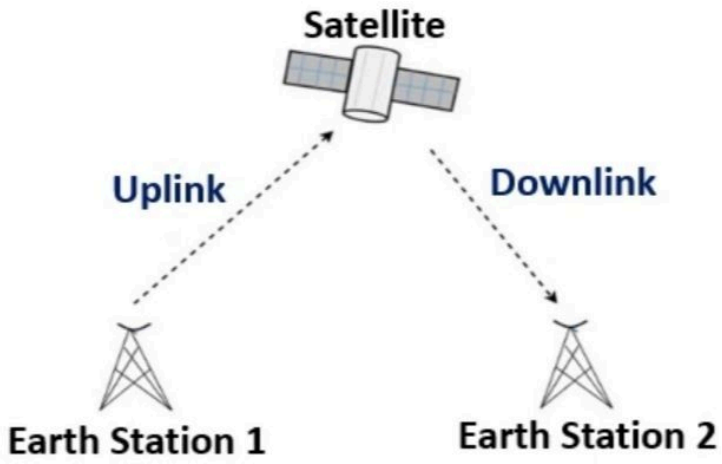
The word "**satellite**" refers to an object that revolves around earth or any other body in space.

Example: moon is a natural satellite. Thousands of artificial or man-made satellites orbit Earth.

What is communication?

The word **"Communication"** refers to the exchange of information between two or more entities, through any medium or channel.

Satellite Communication



If the communication takes place between any two earth stations through a satellite, then it is known as "**satellite communication**".

Introduction to Satellite Communication

What is the need of Satellite Communication?

- Earlier two types of propagations were used for communication i.e. Ground wave propagation and Sky-wave propagation. The maximum distance covered by both of them is limited to 1500 km.
- **Satellite communication** overcomes this limitation. It provides communication for long distances, even which is beyond the line of sight.

- The idea of Satellite Communication was conceived by Arthur C. Clarke a famous British Science fiction writer in year 1945.
- The first Satellite was launched in year 1957.
- Satellites are specifically made for telecommunication purpose.
- They are used for mobile applications such as communication to ships, vehicles, planes, hand-held terminals and for TV and radio broadcasting.

Introduction to Satellite Communication

Communication Frequencies

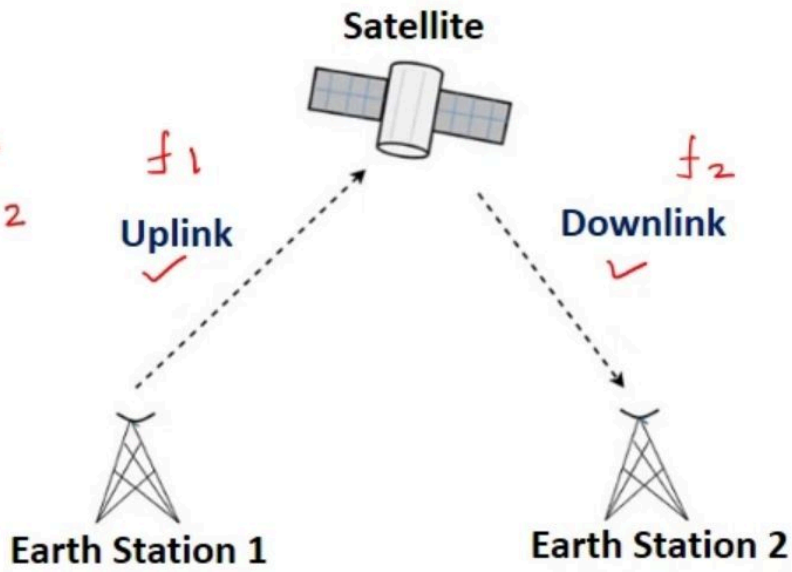
Uplink (Earth to Satellite)

- C Band: around 6 GHz
- K_u Band: around 14 GHz
- K_a Band: around 30 GHz

Downlink (Satellite to Earth)

- C Band: around 4 GHz
- K_u Band: around 12 GHz
- K_a Band: around 20 GHz

6/4 GHz
14/12 GHz
30/20 GHz



Introduction to Satellite Communication

Advantages of Satellite Communication

- Point to multipoint communication is possible
- Area of coverage is more than that of terrestrial systems
- Circuits for the satellite can be installed easily
- The resistance of these circuits is excellent
- With the help of satellite communication, every corner of the earth can be covered
- Satellite cost is independent of the of the distance
- Sending and receiving information is independent of distance
- More bandwidth and broadcasting possibilities
- 24 hours communication can be achieved

Disadvantages of Satellite Communication

- Initial expenditure is high
- There can be congestion of frequencies
- Propagation issues and interference may arise
- Launching of satellites into orbits is an expensive process
- Propagation delay of satellite systems is more than that of conventional terrestrial systems
- Time delay reduces the efficiency of satellite in data transmission etc.
- Difficult to repair if any problem occurs in a satellite system
- Transmitter and receiver used in satellite communication requires high power and large diameter antennas
- Due to ageing effect the efficiency of satellite components decreases
- Free space loss is more

Applications of Satellite Communication

- Radio broadcasting
- TV broadcasting like Direct To Home (DTH)
- Internet applications like GPS
- Military Applications
- Navigations
- Remote Sensing Applications
- Earth Observation
- Weather Condition Monitoring & Forecasting

Frequency Allocation for Satellite

- Allocation of frequencies to satellite services is a complicated process which requires international coordination and planning.
- International Telecommunication Union (ITU) carries out this process.
- To implement this frequency planning, the world is divided into three regions.

WORLD

Division of World for frequency allocation

Then within these regions, frequency bands are allocated to various satellite services

Region-1

- Europe
- Africa
- Mongolia

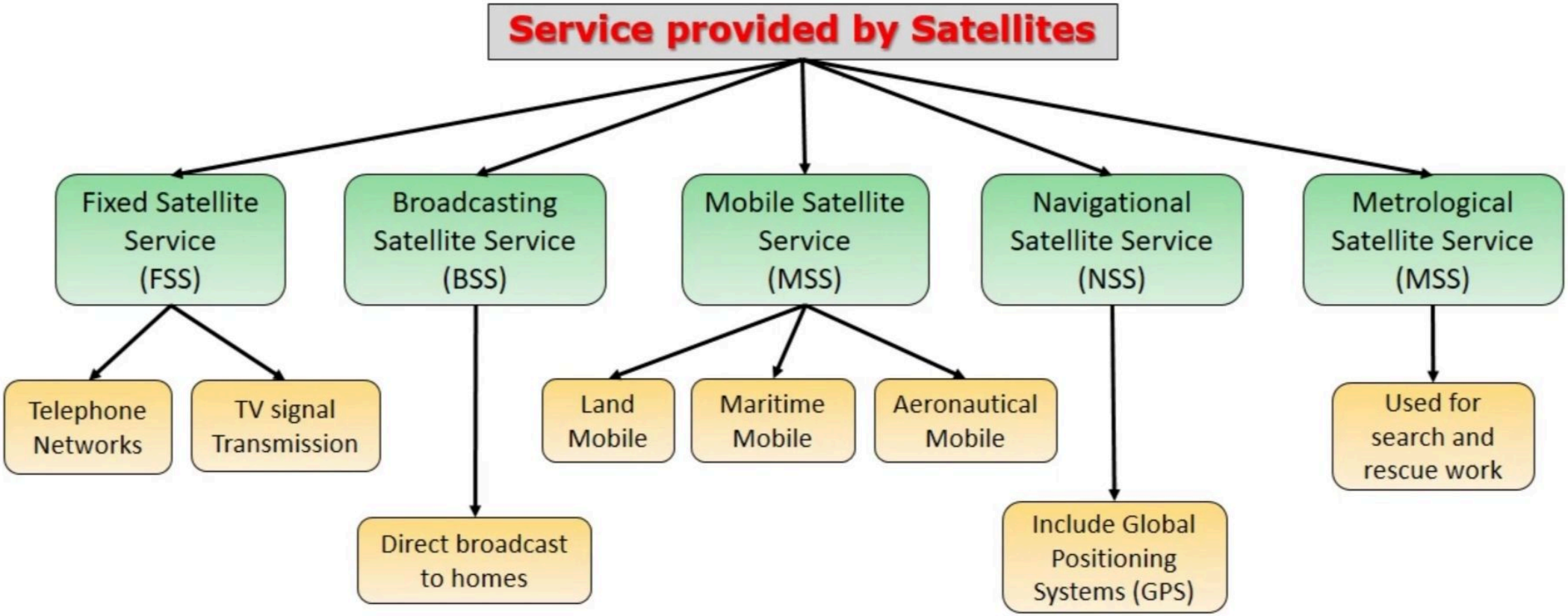
Region-2

- North America
- South America
- Greenland

Region-3

- Asia (excluding region 1 areas)
- Australia
- South West Pacific

Frequency Allocation for Satellite



Frequency Allocation for Satellite

Frequency Band (GHZ) Designations

S. No	Band Designation	Frequency Range (GHz)
1	VHF	0.1-0.3
2	UHF	0.3-1.0
3	L	1-2
4	S	2-4
5	C	4-8
6	X	8-12
7	Ku	12-18
8	K	18-27
9	Ka	27-40
10	VHF	40-75
11	W	75-110
12	mm	110-300
13	µm	300-3000

Frequency Allocation for Satellite

Frequency Band (GHZ) for Satellites Services

S. No	Band Designation	Downlink (GHz)	Uplink (GHz)
1	UHF - Military	250-270 (appx.)	292-312 (appx.)
2	C - Commercial	3700-4200	5925-6425
3	X - Military	7250-7750	7900-8400
4	Ku - Commercial	11700-12200	14000-14500
5	Ka - Commercial	17700-21200	27500-30000
6	Ka - Military	20200-21200	43500-45500

Elements of Satellite Communication

Two major elements of Satellite Communication Systems are:

- Ground Segment
- Space Segment

Space Segment includes:

- Satellite
- Communication Transponder
- Communication Antenna
- Electrical Power System
- Launch Vehicle

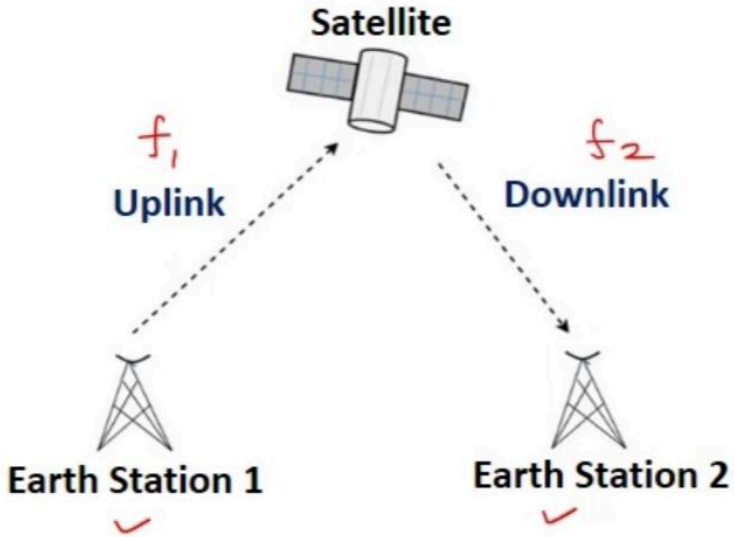
Ground Segment includes:

- Earth Station
- User Terminals and Interface
- Transmit Equipment's
- Receive Equipment's
- Antenna Systems

Elements of Satellite Communication

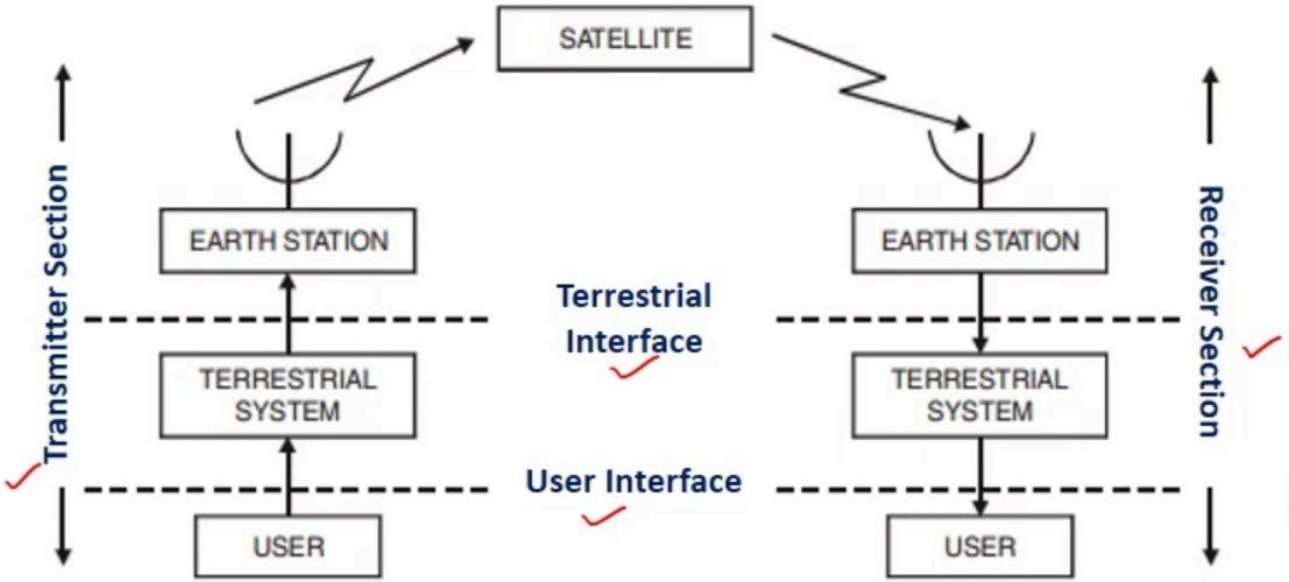
Block Diagram of Satellite Communication System

- An earth station transmits signal to the satellite using a highly directional dish antenna.
- The transmission of signal from earth station 1 to satellite through a channel is known as uplink.
- The transmission of signal from satellite to earth station 2 through a channel is known as downlink.
- Uplink frequency is the frequency at which, earth station 1 is communicating with satellite.
- The satellite transponder converts this signal into another frequency and sends it down to the earth station 2.
- The frequency at which earth station 2 received signal is known as Downlink frequency.



Elements of Satellite Communication

Block Diagram of Satellite Communication System



6/4 GHz

6 GHz → uplink freq

4 GHz → Down link freq.

500 MHz

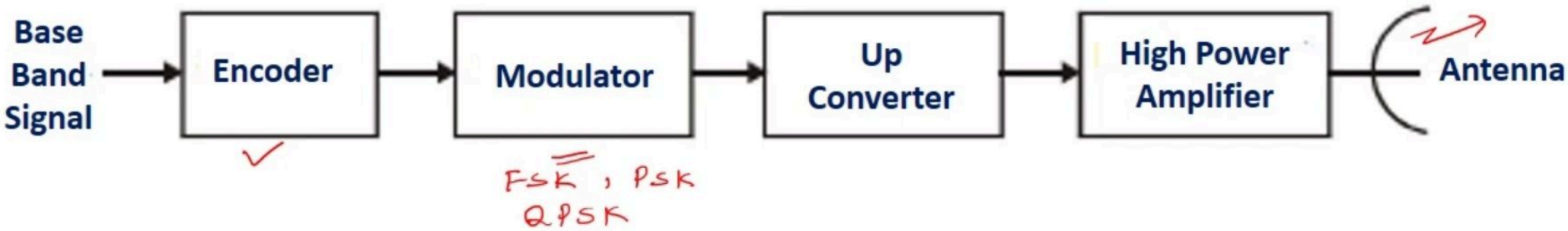
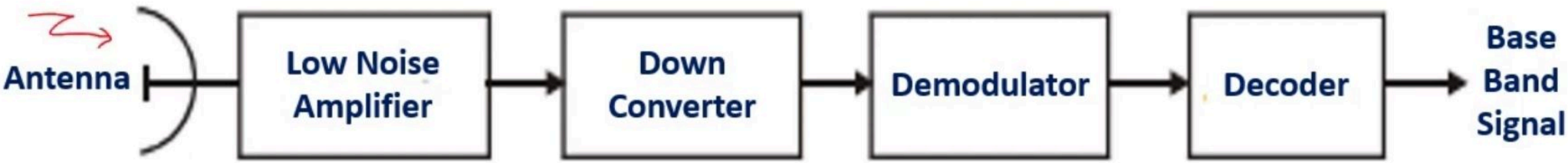
5.725 – 7.075 GHz

3.4 – 4.8 GHz

- User generates a baseband signal which is processed through a terrestrial network.
- Terrestrial network may be a telephone switch or dedicated link to a earth station.
- Repeater is a circuit, which increases the strength of received signal and then transmits it. This repeater works as a transponder.
- It changes the frequency band of the transmitted signal from the received one.
- To avoid interference uplink frequency must be different from downlink frequency.

Elements of Satellite Communication

Earth Station Transmitter



Earth Station Receiver